Personalized Graph Walks for Movie Recommendation

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Assignment: A06 — Graph-Based Recommendations with Personalized PageRank

ABSTRACT

Recommending unseen movies requires distinguishing personal graph structure from global popularity. This study implements TensorFlow.js PageRank and user-seeded PPR on the supplied MovieLens 100K training graph and compares them with popularity and the supplied matrix-factorization reference under identical candidate and user rules. Validation selects link-following probability 0.50, and PPR achieves test macro Recall@10 of 0.11219 over 861 users, versus 0.02114 for the trained MF reference. PPR is the preferred replacement candidate for this baseline, subject to online validation and the substantial limitation that the reference MF collapses to tied scores.

1. Introduction

The task is to rank unseen training-catalog movies for existing users from positive rating links. A user who liked *Star Wars* and several dramas should receive eligible movies informed by those connections; simply returning well-connected movies may ignore that history. The contribution is an implemented and browser-verified PPR pipeline with shared-cohort comparisons, two fixed-user examples and auditable exports. The newer weekly PPR specification [5] governs A06; the older generic schedule's two-tower topic is superseded.

2. Related Work

MovieLens records explicit ratings and timestamps [1,7]. PageRank weights incoming links by their sources' outgoing degree [2]; nonuniform restart biases the stationary distribution [3, §2]. Here the restart is at a user, rather than a topic. TensorFlow.js supplies the actual matrix operations and memory controls [4]. The included MF baseline and its checkpoint rule come from the instructor's source [6]. Popularity is a simple control, global PageRank isolates the effect of personalized restart, and MF provides a learned rating-score alternative; none is assumed superior in advance.

3. Method

Pipeline: load original files → per-user split → graph → browser checks → MF training → validation alpha selection → frozen test evaluation → lists and exports.

Loaded data / split	Training graph / candidates
943 users · 1,682 catalog movies · 100,000 ratings	943 user + 1,611 movie = 2,554 nodes
TRAIN 79,619 · VAL 9,596 · TEST 10,785	45,602 binary positive pairs · 91,204 directed arcs
Sparsity: 93.695331% = 1 − 100000/(943×1682)	1 zero-degree user (685) · 253 zero-degree movies
Per-user timestamp, then movie ID; floors 80%/10%; test remainder	All training-observed IDs; contiguous maps; no padding

Users with fewer than five ratings are skipped (observed: zero). A training rating ≥4 creates one undirected user–movie edge; repeated positive pairs count once. Missing ratings remain unobserved. Low-rated training records still define graph nodes, candidates and exclusions. The 71 catalog movies absent from training are separate from the 253 training candidates with no positive edges.

$$A = [[0, R], [R^T, 0]] \quad \cdot \quad p_0 = v$$
$$p_{next} = \alpha(Q^T p + d\, v) + (1 - \alpha)v$$

Q normalizes nonzero source rows; dangling rows stay zero. The transpose collects incoming mass. Uniform v gives global PageRank; one-hot user v gives PPR. With d the current mass on dangling nodes, sum($Q^T p$)=1−d, so d v restores mass before restart mixing. Alpha is link-following probability. Both methods use this same policy, without post-update renormalization.

Method and measured evaluation

3. Method (continued)

Core propagation uses batched rows (up to 64 seeds), degree-safe user/movie blocks and two `tf.matMul` calls, rather than a square transition tensor. Explicit directed fixtures use pQ. A dense JavaScript Q is allocated only by the diagnostic normalization API. Each seed freezes at its own first L1 change $<10^{-6}$; cap 500 remains nonconverged. Graph/alpha/seed/tolerance/cap caches retain numeric arrays. `tf.ready()` is awaited; tidy/dispose releases temporaries and replaced graph tensors. Synchronous readbacks preserve the starter's synchronous interfaces and can block the UI.

Retrieval: all 1,611 training movies are candidates. VAL removes every train-rated movie; TEST removes train- and validation-rated movies, regardless of rating. Sort scores descending, exact ties by ascending movie ID. Relevance requires held-out rating ≥4, candidate membership and eligibility. Macro Recall@10 averages each eligible user's hits/relevant count. All four methods share each split's cohort: positive training edge, MF score availability and at least one eligible relevant movie. MF is trained on this package, without another homework solution.

4. Experiments

Split	Shared users	No eligible relevant users	No positive edge users	Relevant pairs outside candidates
VAL	846	97	1	4
TEST	861	82	1	13

Exclusion reasons overlap: user 685 is already among the no-eligible users. Total exclusions are 97/82, not 98/83. Eligible relevant pairs are 4,614/5,142 in VAL/TEST; prior-rated filtering removes zero held-out relevant pairs in these unique-pair files. MF adds no scoring exclusions.

PPR alpha	VAL Recall@10 (846)	Updates, min–max	Capped / 846
0.50	0.11448316	21–22	0
0.70	0.10479695	41–42	0
0.85	0.09002077	88–95	0
0.95	0.07375030	272–500	27

Select the highest unrounded validation recall; exact ties favor smaller alpha. Selected **α=0.50** is persisted before TEST and also controls global PageRank; global alpha is not separately optimized. At α=.95, 27 validation walks reach the cap (worst residual 1.823×10^{-6}); that row is a capped approximation. No tolerance was relaxed, and no test result selected a setting.

Method	TEST Recall@10 (861)	Mean list popularity	Convergence at α=.50
popularity	0.07602291	271.997	not applicable
pagerank	0.07969324	257.820	19 updates; 5.436×10^{-7}
ppr	0.11218853	244.207	21–22; max 9.994×10^{-7}
mf	0.02113749	96.761	not applicable

MF: 16 factors, seeded normal initialization SD .05, mulberry32 seed 20261013, 30 epochs, Adam lr .01, L2 .05 on both entire factor matrices each batch, batch 512. Earliest minimum validation-RMSE checkpoint: epoch 1, RMSE 1.18299443. Scores are TRAIN mean + dot product, unclipped, as supplied [6]; MF's validation objective differs from PPR's. TRAIN mean is 3.580602620. The fixed users' MF lists contain tied float32 scores, so this is a weak reference, not evidence against all MF models.

Environment and measured time: Chrome 156.0.8078.48, macOS 26.5, Apple M5, 16 GiB; TensorFlow.js 4.22.0, d3 7.9.0, WebGL via ANGLE/Metal. MF training took 14.412s, validation sweep 21.119s, and first four-method TEST comparison 9.588s. Scope includes readback/ranking; TEST reuses validation walks when applicable. Timings describe this device, not algorithm speed across devices.

Reference restart at .85	Expected stationary p	Observed tensor p
uniform	0.3725269, 0.1958239, 0.3941492, 0.0375000	0.3725269, 0.1958238, 0.3941493, 0.0375000
A only	0.4522329, 0.1921990, 0.3555681, 0.0000000	0.4522330, 0.1921991, 0.3555682, 0.0000000

Executed verification: 12 supplied + 15 additional browser checks pass; zero FAIL/PENDING. Directed vectors match within 10^{-6}; separate dangling fixture gives [0.3508772, 0.6491228]. Full diagnostic row error $<10^{-12}$; all 254 dangling rows are zero. Sparse/raw-ID seeds, low-rating exclusions, ties, unavailable MF, cap behavior and macro example (½+1)/2=.75 pass. Maximum cohort mass drift is 1.431×10^{-6} in float32. After MF warmup, repeated uncached fixture runs and graph rebuilds stay at 6 live tensors; restoring releases the blocks, leaving 2 MF variables. Independent input audits are separate from these JavaScript tests.

4. Experiments · user 1 and all four Top-10 lists

Fixed user A = 1. Selected as one of the two smallest TEST-cohort IDs, before comparing their lists. TRAIN contains 217 ratings, of which 131 are positive. Positive-history genre counts include Drama 61, Comedy 39, Action 36; multi-genre movies contribute to each genre.

Training examples: **Star Wars** (50; Action/Adventure/Romance/Sci-Fi/War), **Mr. Holland's Opus** (15; Drama) and **Three Colors: Red** (59; Drama), each rated 5. These examples support a broad history rather than a single genre.

All lists below use TEST exclusions. Graph methods use the primary validation-selected α=.50. Popularity is distinct positive TRAIN users per movie. Graph score columns display probability ×1,000; divide by 1,000 to recover the walk probability. Scores are rounded only for presentation.

popularity · mean popularity 246.8

#	Movie ID · title / genres	Score	Pop.
1	**100** · Fargo (1996) Cri/Dra/Thr	377	377
2	**258** · Contact (1997) Dra/SF	307	307
3	**286** · English Patient, The (1996) Dra/Rom/War	279	279
4	**300** · Air Force One (1997) Act/Thr	235	235
5	**288** · Scream (1996) Hor/Thr	226	226
6	**318** · Schindler's List (1993) Dra/War	224	224
7	**313** · Titanic (1997) Act/Dra/Rom	219	219
8	**302** · L.A. Confidential (1997) Cri/Noir/Mys/Thr	202	202
9	**357** · One Flew Over the Cuckoo's Nest (1975) Dra	202	202
10	**483** · Casablanca (1942) Dra/Rom/War	197	197

pagerank · mean popularity 239.4

#	Movie ID · title / genres	Score ×10³	Pop.
1	**258** · Contact (1997) Dra/SF	2.375186	307
2	**286** · English Patient, The (1996) Dra/Rom/War	2.291253	279
3	**100** · Fargo (1996) Cri/Dra/Thr	2.271696	377
4	**313** · Titanic (1997) Act/Dra/Rom	2.063426	219
5	**300** · Air Force One (1997) Act/Thr	2.003419	235
6	**288** · Scream (1996) Hor/Thr	1.868373	226
7	**302** · L.A. Confidential (1997) Cri/Noir/Mys/Thr	1.798134	202
8	**294** · Liar Liar (1997) Com	1.488346	182
9	**275** · Sense and Sensibility (1995) Dra/Rom	1.215865	178
10	**222** · Star Trek: First Contact (1996) Act/Adv/SF	1.192817	189

ppr · mean popularity 228.4

#	Movie ID · title / genres	Score ×10³	Pop.
1	**100** · Fargo (1996) Cri/Dra/Thr	0.685519	377
2	**258** · Contact (1997) Dra/SF	0.439866	307
3	**286** · English Patient, The (1996) Dra/Rom/War	0.431018	279
4	**318** · Schindler's List (1993) Dra/War	0.403817	224
5	**357** · One Flew Over the Cuckoo's Nest (1975) Dra	0.377166	202
6	**483** · Casablanca (1942) Dra/Rom/War	0.359689	197
7	**222** · Star Trek: First Contact (1996) Act/Adv/SF	0.332785	189
8	**475** · Trainspotting (1996) Dra	0.328090	164
9	**275** · Sense and Sensibility (1995) Dra/Rom	0.320185	178
10	**423** · E.T. the Extra-Terrestrial (1982) Child/Dra/Fan/SF	0.305668	167

mf · mean popularity 72.2

#	Movie ID · title / genres	Score	Pop.
1	**5** · Copycat (1995) Cri/Dra/Thr	3.58060265	28
2	**6** · Shanghai Triad (Yao a yao yao dao waipo qiao) (1995) Dra	3.58060265	12
3	**18** · White Balloon, The (1995) Dra	3.58060265	1
4	**20** · Angels and Insects (1995) Dra/Rom	3.58060265	31
5	**32** · Crumb (1994) Doc	3.58060265	46
6	**74** · Faster Pussycat! Kill! Kill! (1965) Act/Com/Dra	3.58060265	2
7	**100** · Fargo (1996) Cri/Dra/Thr	3.58060265	377
8	**102** · Aristocats, The (1970) Ani/Child	3.58060265	18
9	**111** · Truth About Cats & Dogs, The (1996) Com/Rom	3.58060265	131
10	**129** · Bound (1996) Cri/Dra/Rom/Thr	3.58060265	76

Genre key: Act Action · Adv Adventure · Ani Animation · Child Children's · Com Comedy · Cri Crime · Doc Documentary · Dra Drama · Fan Fantasy · Noir Film-Noir · Hor Horror · Mys Mystery · Rom Romance · SF Sci-Fi · Thr Thriller · War War.

Each list has 10 eligible movies and no zero scores. MF scores tie exactly; movie ID breaks ties after filtering.

4. Experiments · user 2 and all four Top-10 lists

Fixed user B = 2. Selected as one of the two smallest TEST-cohort IDs, before comparing their lists. TRAIN contains 49 ratings, of which 32 are positive. Positive-history genre counts include Drama 18, Romance 11, Comedy 11; multi-genre movies contribute to each genre.

Training examples: **Fargo** (100; Crime/Drama/Thriller), **Sense and Sensibility** (275; Drama/Romance), **Emma** (283; Drama/Romance) and **Star Wars** (50), each rated 5. All these seen movies are excluded for this user.

All lists below use TEST exclusions. Graph methods use the primary validation-selected α=.50. Popularity is distinct positive TRAIN users per movie. Graph score columns display probability ×1,000; divide by 1,000 to recover the walk probability. Scores are rounded only for presentation.

popularity · mean popularity 262.1

#	Movie ID · title / genres	Score	Pop.
1	**181** · Return of the Jedi (1983) Act/Adv/Rom/SF/War	339	339
2	**174** · Raiders of the Lost Ark (1981) Act/Adv	312	312
3	**98** · Silence of the Lambs, The (1991) Dra/Thr	308	308
4	**56** · Pulp Fiction (1994) Cri/Dra	263	263
5	**172** · Empire Strikes Back, The (1980) Act/Adv/Dra/Rom/SF/War	261	261
6	**7** · Twelve Monkeys (1995) Dra/SF	234	234
7	**79** · Fugitive, The (1993) Act/Thr	230	230
8	**64** · Shawshank Redemption, The (1994) Dra	228	228
9	**318** · Schindler's List (1993) Dra/War	224	224
10	**173** · Princess Bride, The (1987) Act/Adv/Com/Rom	222	222

pagerank · mean popularity 247.4

#	Movie ID · title / genres	Score ×10³	Pop.
1	**181** · Return of the Jedi (1983) Act/Adv/Rom/SF/War	1.988940	339
2	**174** · Raiders of the Lost Ark (1981) Act/Adv	1.552726	312
3	**98** · Silence of the Lambs, The (1991) Dra/Thr	1.525544	308
4	**7** · Twelve Monkeys (1995) Dra/SF	1.373775	234
5	**56** · Pulp Fiction (1994) Cri/Dra	1.372618	263
6	**172** · Empire Strikes Back, The (1980) Act/Adv/Dra/Rom/SF/War	1.318299	261
7	**117** · Rock, The (1996) Act/Adv/Thr	1.308004	215
8	**121** · Independence Day (ID4) (1996) Act/SF/War	1.297289	207
9	**268** · Chasing Amy (1997) Dra/Rom	1.265609	146
10	**222** · Star Trek: First Contact (1996) Act/Adv/SF	1.192817	189

ppr · mean popularity 226.7

#	Movie ID · title / genres	Score ×10³	Pop.
1	**181** · Return of the Jedi (1983) Act/Adv/Rom/SF/War	0.697339	339
2	**9** · Dead Man Walking (1995) Dra	0.497664	182
3	**7** · Twelve Monkeys (1995) Dra/SF	0.474374	234
4	**117** · Rock, The (1996) Act/Adv/Thr	0.456362	215
5	**98** · Silence of the Lambs, The (1991) Dra/Thr	0.446510	308
6	**121** · Independence Day (ID4) (1996) Act/SF/War	0.430713	207
7	**174** · Raiders of the Lost Ark (1981) Act/Adv	0.402851	312
8	**268** · Chasing Amy (1997) Dra/Rom	0.401811	146
9	**475** · Trainspotting (1996) Dra	0.401605	164
10	**15** · Mr. Holland's Opus (1995) Dra	0.400918	160

mf · mean popularity 111.4

#	Movie ID · title / genres	Score	Pop.
1	**2** · GoldenEye (1995) Act/Adv/Thr	3.58060265	41
2	**3** · Four Rooms (1995) Thr	3.58060265	27
3	**4** · Get Shorty (1995) Act/Com/Dra	3.58060265	96
4	**5** · Copycat (1995) Cri/Dra/Thr	3.58060265	28
5	**6** · Shanghai Triad (Yao a yao yao dao waipo qiao) (1995) Dra	3.58060265	12
6	**7** · Twelve Monkeys (1995) Dra/SF	3.58060265	234
7	**8** · Babe (1995) Child/Com/Dra	3.58060265	137
8	**9** · Dead Man Walking (1995) Dra	3.58060265	182
9	**11** · Seven (Se7en) (1995) Cri/Thr	3.58060265	152
10	**12** · Usual Suspects, The (1995) Cri/Thr	3.58060265	205

Genre key: Act Action · Adv Adventure · Ani Animation · Child Children's · Com Comedy · Cri Crime · Doc Documentary · Dra Drama · Fan Fantasy · Noir Film-Noir · Hor Horror · Mys Mystery · Rom Romance · SF Sci-Fi · Thr Thriller · War War.

Each list has 10 eligible movies and no zero scores. MF scores tie exactly; movie ID breaks ties after filtering.

Controlled evidence, decision and disclosure

Method, primary setting	Mean popularity A / B	A–B shared movies
popularity	246.8 / 262.1	1
pagerank	239.4 / 247.4	1
ppr	228.4 / 226.7	1
mf	72.2 / 111.4	2

PPR one-factor setting	Ordered TEST Top-10 movie IDs	Mean popularity	Overlap vs baseline
baseline: user 1, α=0.85	100, 258, 286, 318, 357, 483, 288, 300, 222, 313	245.5	10
change seed only: user 2, α=0.85	181, 98, 174, 7, 56, 117, 172, 9, 121, 64	254.9	0
change alpha only: user 1, α=0.50	100, 258, 286, 318, 357, 483, 222, 475, 275, 423	228.4	7

Changing only A→B at .85 yields no shared movies, with mean popularity 245.5 vs 254.9. User-specific exclusions also change, so the displayed difference is not solely a score effect. Changing only A's alpha .85→.50 keeps seven movies: *Trainspotting*, *Sense and Sensibility* and *E.T.* replace *Scream*, *Air Force One* and *Titanic*; mean popularity falls 245.5→228.4. The alpha-only .50 list is exactly the primary user A PPR list on page 3. These changes establish sensitivity, not better discovery or current intent.

5. Discussion

Business decision — replace this MF with PPR as the main-recommender candidate, subject to online testing. PPR has the highest common-cohort Recall (.11219 vs .02114 for this MF and .07602 for popularity). The A/B histories and complete lists show plausible drama and science-fiction connections; their primary PPR means (228.4/226.7) are lower than popularity's (246.8/262.1), but overlap is one movie for both methods. Lower popularity or different lists alone does not establish relevance. PPR scores are graph-walk probabilities, not predicted ratings or purchase probabilities.

Observed limits. MF's selected weights are tiny (maximum absolute user/item factors 4.680×10^{-5}/5.505×10^{-5}); its displayed scores collapse to the training mean at float32 precision. This is consistent with the supplied whole-matrix L2 penalty, but no causal regularization ablation was run. The baseline definition was preserved instead of tuning it after inspecting test results. The alpha .95 caps and synchronous UI blocking are additional limits. Next, compare a validation-tuned MF regularization setting with PPR on a new untouched temporal holdout.

Real failure and correction. The independent export audit raised `AssertionError: ('cohort', 'test|popularity|0.85')`: a tiny retrieval test had reused the mutable report dictionary and left a synthetic cohort in it. Giving the fixture its own report store and resetting that store before experiments removed contamination. A clean browser rerun and independent audit then verified all 6,828 per-user metric rows and 110 recommendation rows. Earlier failure logs are preserved; the primary metric values did not change. A separate added reference check initially used the wrong helper shape and was corrected transparently.

What worked / what remains uncertain. Directed fixtures catch orientation errors, sparse-ID tests catch shifted seeds, and a shared population makes the four-method comparison interpretable. The weak MF and slower convergence near .95 show why training success and reaching a cap are not quality guarantees. The split is chronological only within users, not globally time-consistent; rating timestamps are not viewing sessions. This warm-user offline experiment does not establish cold-start performance, current-session intent, completion, retention or revenue, and no significance test was run.

Proposed online test. Randomly assign eligible users persistently by user-ID hash to the same main-section layout with supplied MF (control) or PPR .50 (treatment). Primary outcome: completed recommended-movie views per assigned eligible user over seven days. Guardrails: seven-day return rate and p95 serving latency. Keep assignment across sessions/devices and analyze by assigned user to avoid cross-arm contamination. Any improvement remains a hypothesis.

6. AI Usage Disclosure

AI tools used. I used Codex 0.159.2 with model `gpt-6.1-sol` for implementation, debugging, experiments, report drafting and review.

How AI was used. I used AI to implement the seven required functions, develop verification scripts, run experiments, diagnose failures and draft the report.

What I personally verified. I personally reran the code, reviewed the logs and checked the cited sources. I verified $p_1(B)=.85\times(.25/2)+.15\times.25=.14375$, dangling-mass redistribution through v, and TRAIN/VAL exclusion from TEST recommendations, including user 1's movie 260, rated 1. I confirmed validation-only alpha selection and report/export agreement. All 27 browser checks passed; the repeat run reproduced all four test metrics and all eight primary Top-10 lists. I checked the MF score ties and interpreted the comparison as evidence against this baseline, without generalizing to all MF models.

TF.js API claims. I matched `tf.matMul`, `tf.ready`, `tf.tidy`, `dataSync` and `dispose` in the implementation to the TensorFlow.js 4.22.0 API documentation.

References checked. I checked bibliographic details and algorithm passages in Brin–Page §2.1 and Haveliwala §2. Haveliwala's alpha denotes restart probability; here it denotes link-following probability.

Session log reference. My authentic export, `rollout-2026-10-29T16-45-25-01a187e8-8719-7b52-8221-ec6006d41450.jsonl`, records AI-assisted work and my initial checks. I performed my subsequent rerun, log review and source checks outside that session.

Remaining limitations. I did not evaluate online business effects or reproducibility on other devices.

References

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Source verification and provenance

The agent checked algorithm passages in Brin–Page §2.1 and Haveliwala §2 (whose alpha denotes restart, unlike this assignment’s link-following alpha). Dataset counts/format and license were checked against the original GroupLens README. Harper–Konstan authors/year/title were checked against the author-hosted manuscript and registered DOI metadata; the manuscript contains placeholder publication fields, so those placeholders were not copied. TensorFlow.js API behavior was checked against version 4.22.0 documentation. The pinned course source establishes the supplied MF configuration and scoring rule. These initial source checks informed the implementation; my subsequent personal review is described in §6.

Local reproducibility record

Final measured run: `a997e655-c6ff-4d2b-bd0e-23dcfa6e4287`. Completed 2026-10-29T13:58:47.374198+00:00 UTC. Source/input SHA-256 hashes, environment, stage times, checkpoint weights, full per-user recalls and per-seed convergence are retained in the local evidence. Input data, Week 5 files, supplied PDFs and upstream MIT attribution were preserved. The PDF contains the method, measured comparisons, verification, all eight primary lists, limitations and disclosure so assessment does not require those local files.

This report has five body pages plus this references page.

Conversation and session files

Public repository: https://github.com/agiccoder/hw6-ppr-submission
Download session JSON · Download authentic native JSONL
The JSON copy preserves all 362 events in their original order; the native JSONL export is unchanged. Both links identify the fixed session-file commit.